

TA(s) Consulted: [Name(s) of TA(s) you spoke to]

Instrument Cataloguer

1 Team Members

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2 Task Overview

We are creating a program to catalogue various orchestral instruments based on their waveform. Many musicians find it very difficult to catalogue instruments because many instruments do not have physical identifiers on them.

This task is challenging because the raw waveform must be converted with a Fourier transform into time–frequency features such as a spectrogram.

3 Task Definition

The data we are working with is an audio .mp3 file. This is a single-label, multi-class classification task with 19 classes, one per instrument in the dataset: flute, oboe, cor anglais (English horn), clarinet, bass clarinet, bassoon, contrabassoon, saxophone, French horn, trombone, trumpet, tuba, banjo, cello, double bass, guitar, mandolin, viola, violin. Each clip is one note by one instrument. We have multiple features: fundamental frequencies, timbre, resonance, harmonics, noisiness/dynamics.

4 Problem Description and Impact

Suppose a recording of a composition uses a certain wind/string instrument, and we want to find the same instrument again. Being able to say what instrument a sound came from based on how it sounds makes it easier to search.

The challenges are: (1) audio analysis requires knowledge of signal processing; we will be using Fourier transforms; (2) the same instrument sounds different across pitches, dynamics and articulation, so the model must not just learn “pitch” or “loudness” as a shortcut; (3) many instruments in the same family share very similar timbre; (4) because the recordings are really clean, this may not transfer to real-world recordings.

5 Data Source(s) and Collection Plan

- Links to the data source(s):
- <https://philharmonia.co.uk/resources/sound-samples/>
- Metadata: each file's name has the instrument, pitch, note duration, dynamics and articulation
- Data is already labeled
- If not collected by you: details on how it was originally collected/annotated/preprocessed
- We will be using a subset of the full database (brass, strings, woodwinds). This is because percussion instruments do not have that many recordings.

6 Dataset Size and Example Data Points

The collection of instruments has 14,226 clips (one note per file).

Example Data Points

1. **Input:** [flute_A4_1_forte_normal.mp3 (a 1-second flute note A4, forte)] **Label:** [flute]
2. **Input:** [clarinet_C4_1_piano_normal.mp3 (clarinet note, C4, piano)] **Label:** [clarinet]
3. **Input:** bassoon_G2_1_fortissimo_normal.mp3 (low bassoon note, G2) **Label:** [bassoon]

7 Proposed Solution

Our pipeline will be as follows: decode the .mp3, convert to mono, resample to 16kHz, trim silences, and normalize clip length by padding or cropping to a fixed length (1-2 seconds, will be determined experimentally).

We can then use the cleaned data to process inputs for a model. We have two options, building hand crafted features like MFCC mean and standard deviation, spectral centroid, rolloff, flatness, zero-crossing rate, and attack time for a classical SVM or random forest model, or log-mel spectrogram for a small CNN with about 4 conv blocks.

We will also evaluate the dataset further:

- A grouped split, where we group all clips by '(instrument, pitch)', so the same note at different dynamics or durations never appears in both train and test. Without this the accuracy of the model would be artificially inflated by memorizing the pitch of the note.
- We use macro F1 score and balanced accuracy as our metrics, since the dataset is imbalanced across instruments. (There are 1,502 violin clips, and while some of the other instruments don't even have clip counts above 100.)
- Tuba, contrabassoon and double bass can be identified by pitch alone. Therefore, we will evaluate on the pitch range where instruments overlap, or hold out whole pitches.

Old stuff.

- Each clip is resampled, trimmed, normalized, then converted to a spectrogram. The target label is one of the 19 instruments.
- What model(s) will you try (no simple linear regression)?
- Instrument recognition is well studied. Our project applies these ideas to a focused instrument subset.
- How will you evaluate whether your model is good (metrics, validation strategy)?
- librosa, torchaudio/PyTorch, scikit-learn, pandas, NumPy, and matplotlib

Sources of Inspiration

1. J. Engel et al., “Neural Audio Synthesis of Musical Notes with WaveNet Autoencoders” (the NSynth dataset), ICML 2017.
2. J. J. Bosch, J. Janer, F. Fuhrmann, and P. Herrera, “A Comparison of Sound Segregation Techniques for Predominant Instrument Recognition in Musical Audio Signals” (the IRMAS dataset), ISMIR 2012.
3. Y. Han, J. Kim, and K. Lee, “Deep Convolutional Neural Networks for Predominant Instrument Recognition in Polyphonic Music,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2017.
4. Q. Kong et al., “PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2020.

8 TA Consultation Summary

TA: Abdulazeer ”Fourier transforms seem difficult, but if you are interested in vision models, then this would be really interesting.”

”You should try to figure out how to measure differences in the signals you get from your fourier transforms/log-mel spectrogram.”

”Do your analysis on classes individually, so that the accuracy doesn’t get inflated by e.g. that violin class ”